completion of the surgery the blood vessel is connected to normal the circulation and the blood flows freely.

Heart Transplantation

Each year over several million people worldwide die of heart disease, than from other conditions. For some patients heart transplant is the only hope.

A heart transplant is a surgical transplantation procedure which is done to replace a diseased or a damaged heart. This procedure is performed on a patient with end stage heart failure or severe coronary artery disease, when other medical ailments or surgical treatments have failed. The most common procedure is to take a functioning heart from a brain dead person (organ donor) and is transplanted in a person with a damaged heart. After the heart transplant the average life span of the person increases.

Cardio Pulmonary Resuscitation (CPR)

YWHGF

In 1956, **James Elam** and **Peter Safar** were the first to use **mouth to mouth resuscitation**. CPR is a life saving procedure that is done at the time of emergency conditions when a person's breath or heart beat has stopped abruptly in case of drowning, electric shock or heart attack. CPR includes rescue of breath, which is achieved by mouth to mouth breathing, to deliver oxygen to the victim's lungs by external chest compressions which helps to circulate blood to the vital organs. CPR must be performed within 4 to 6 minutes after cessation of breath to prevent brain damage or death. Along with CPR, defibrillation is also done. Defibrillation means a brief electric shock is given to the heart to recover the function of the heart.



First heart transplantation was performed in the year 1959. Human heart transplant was performed by Prof. Christian Bernard in South Africa in the year 1967, December 3 at Groote Schuur Hospital, Cape Town. Dr Anangipalli Venugopal was the first to perform heart transplant at AIIMS, India on August 3, 1994.



XI Std Zoology Chapter-7 EM.indd 129 30/12/2022 13:45:53